Generic Convection-Diffusion Transport

Aim: To compare scalar transport in forced flat-plate, forced channel, and natural-convection surrogate velocity fields.

Case definition and parameters: Simulations were analyzed on the following parameters.

Parameter	Value
Coordinate system	Cartesian
Grid points, nx	65
Grid points, ny	65
Raynold's number	100,400
U_lid	2
CFL	0.8
Tolerance	1e-0.8
Max_steps	2000
SOR	1.7

Governing Equation:

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\mathbf{u}\phi) = \nabla \cdot (\Gamma \nabla \phi) + S_\phi,$$

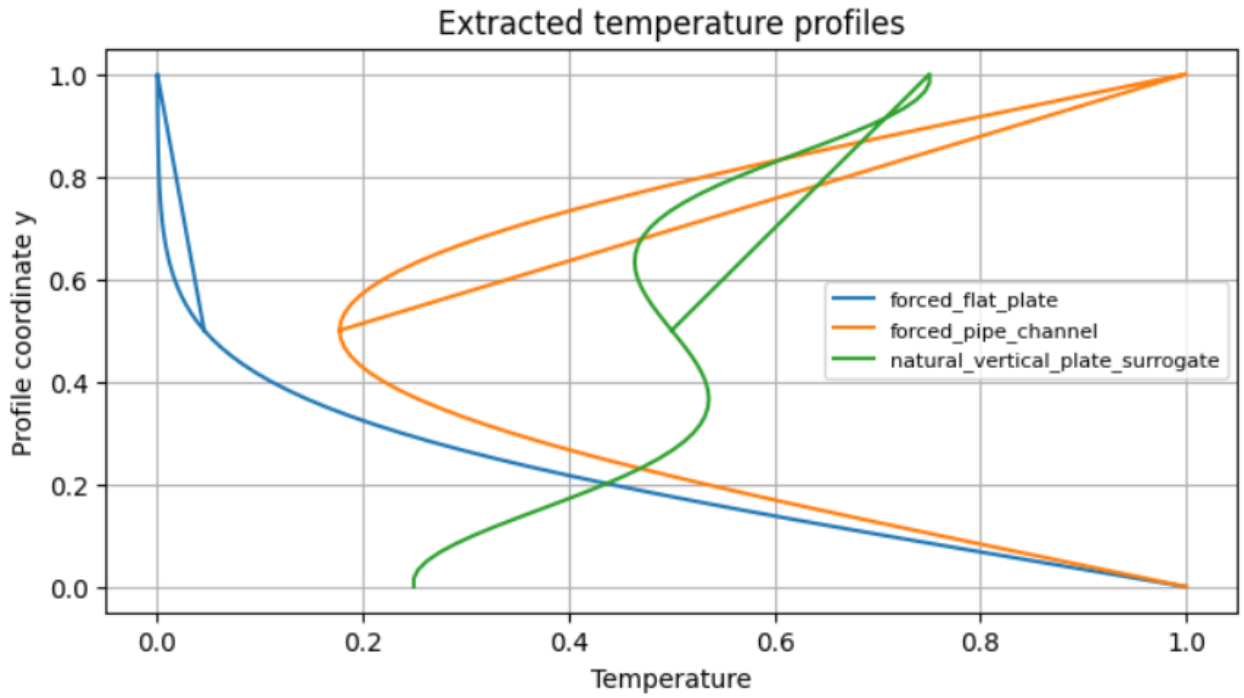
with cell Peclet number $Pe = |\mathbf{u}|\Delta x/\Gamma$.

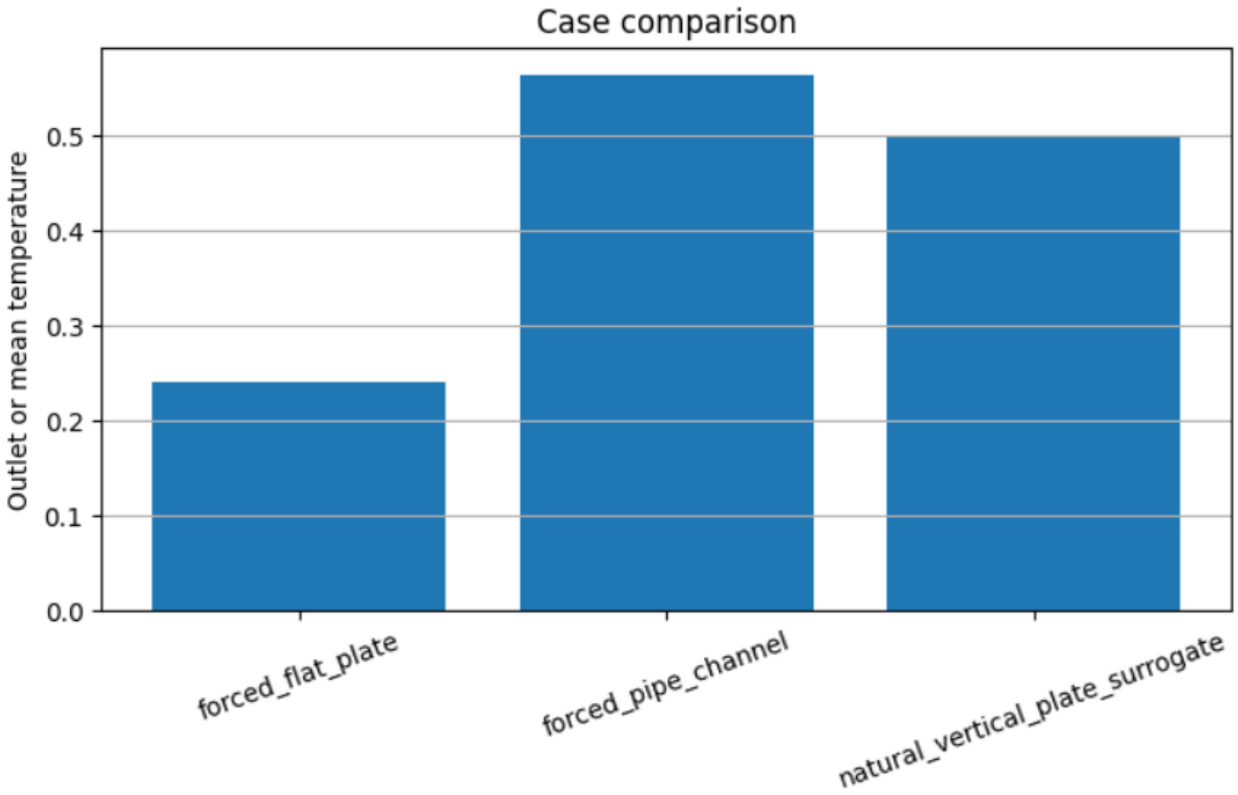
Boundary Conditions: Case-dependent thermal boundaries: heated flat plate, heated channel walls, or hot/cold vertical walls for the recirculating surrogate.

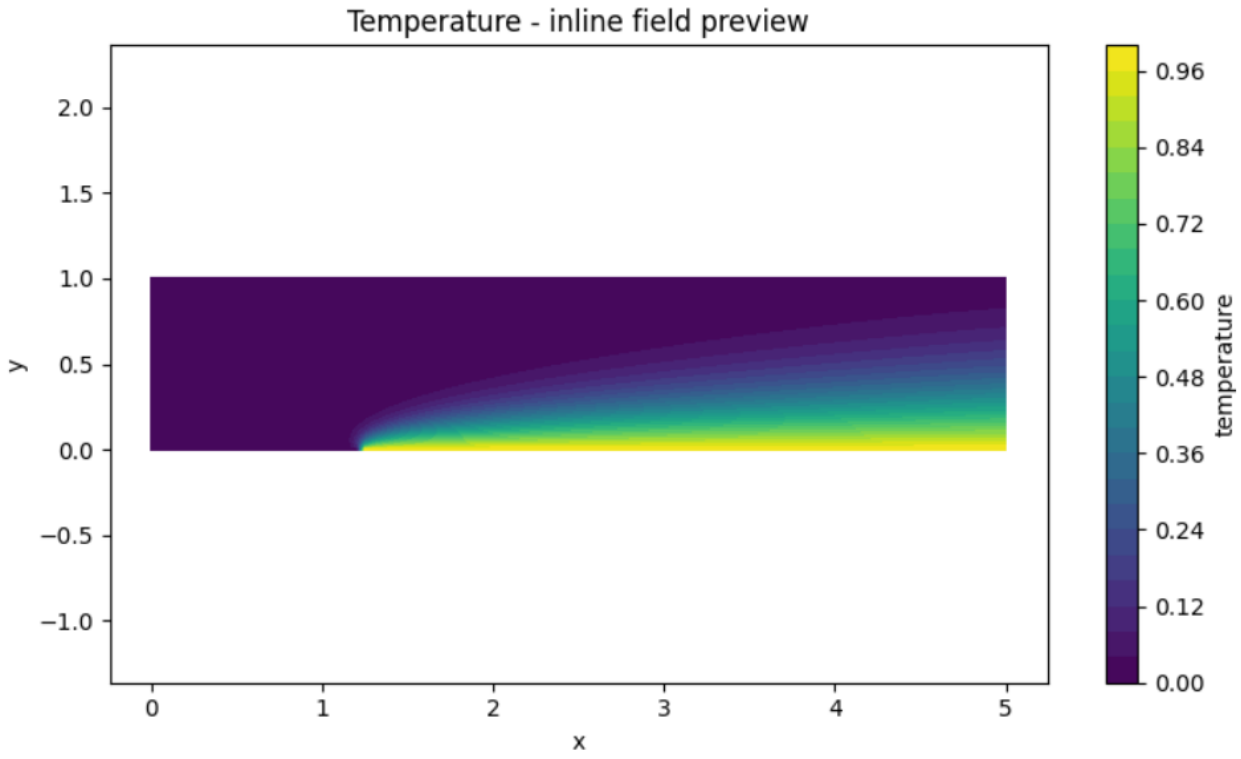
Numerical Method: Finite-volume-style transient iteration with upwind convection and central diffusion for three prescribed velocity fields.

Observations: Residual, outlet or mean temperature, average wall flux, scalar boundary-layer thickness, local Peclet number, and field shape at Peclet numbers 50, 200, and 1000. Compare field shape, boundary-layer thickness, wall flux, outlet or mean temperature, and convergence.

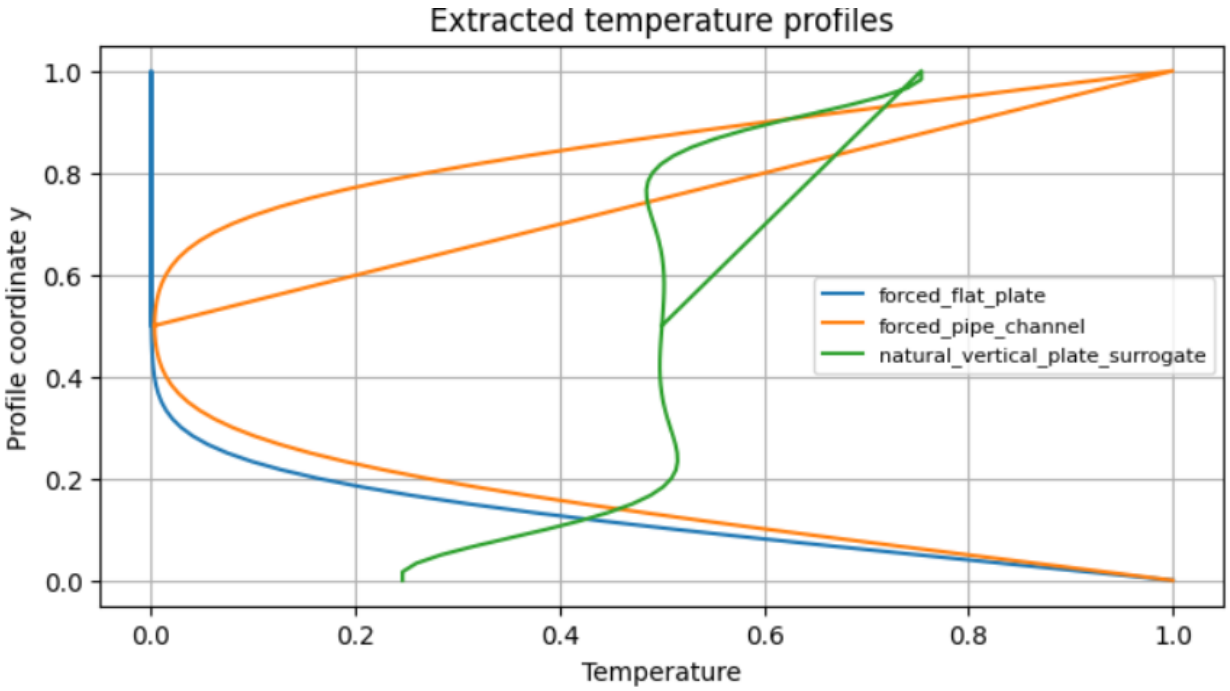
For Peclet number 50:

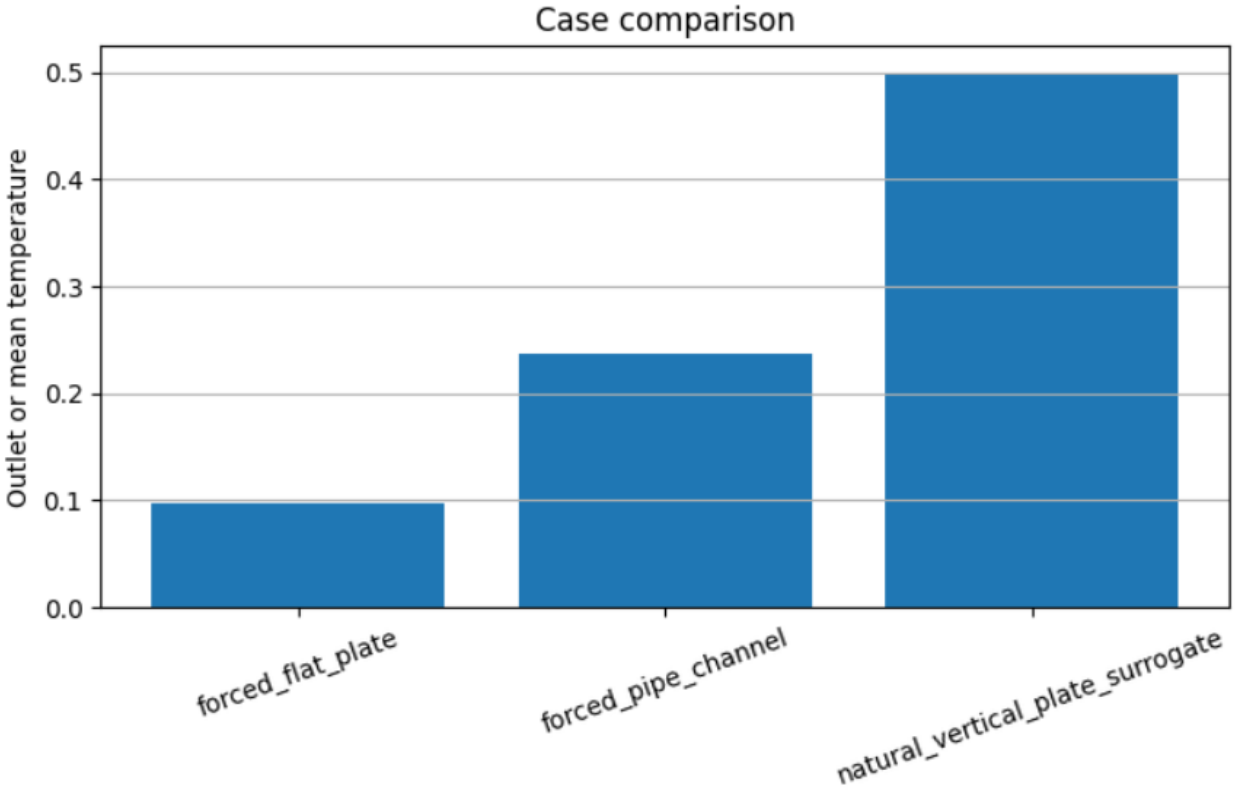


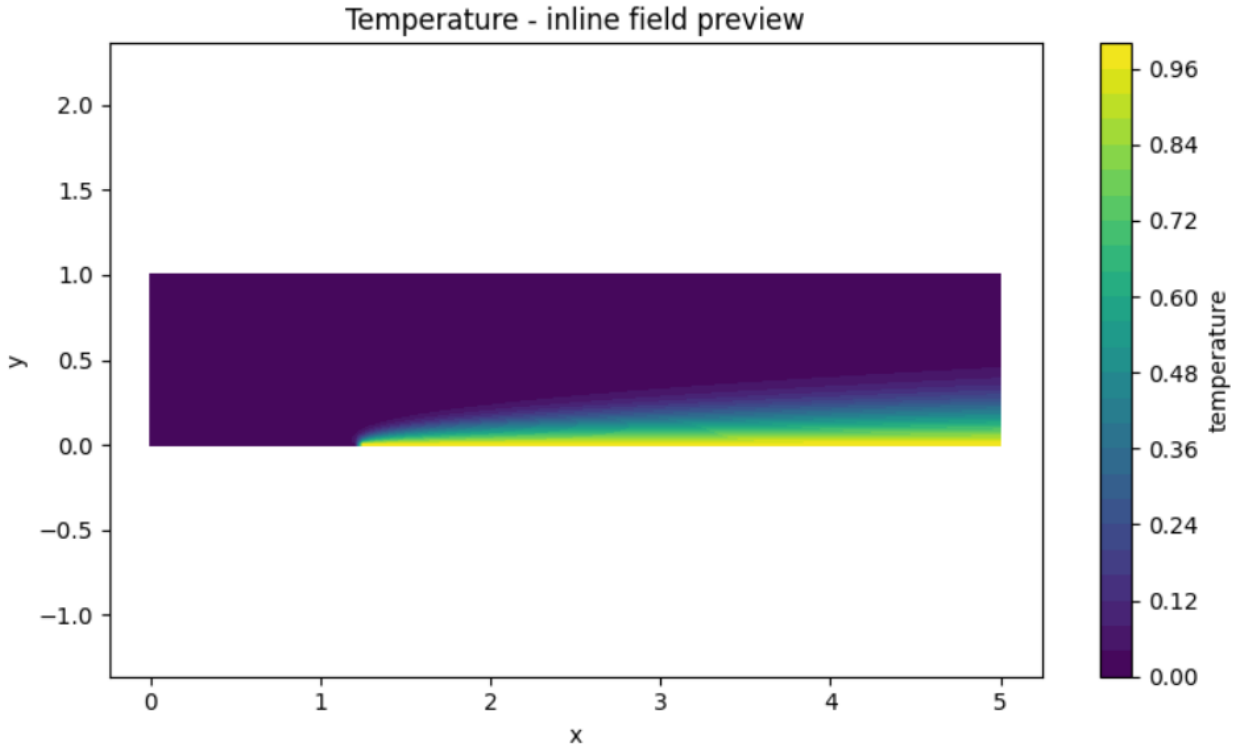




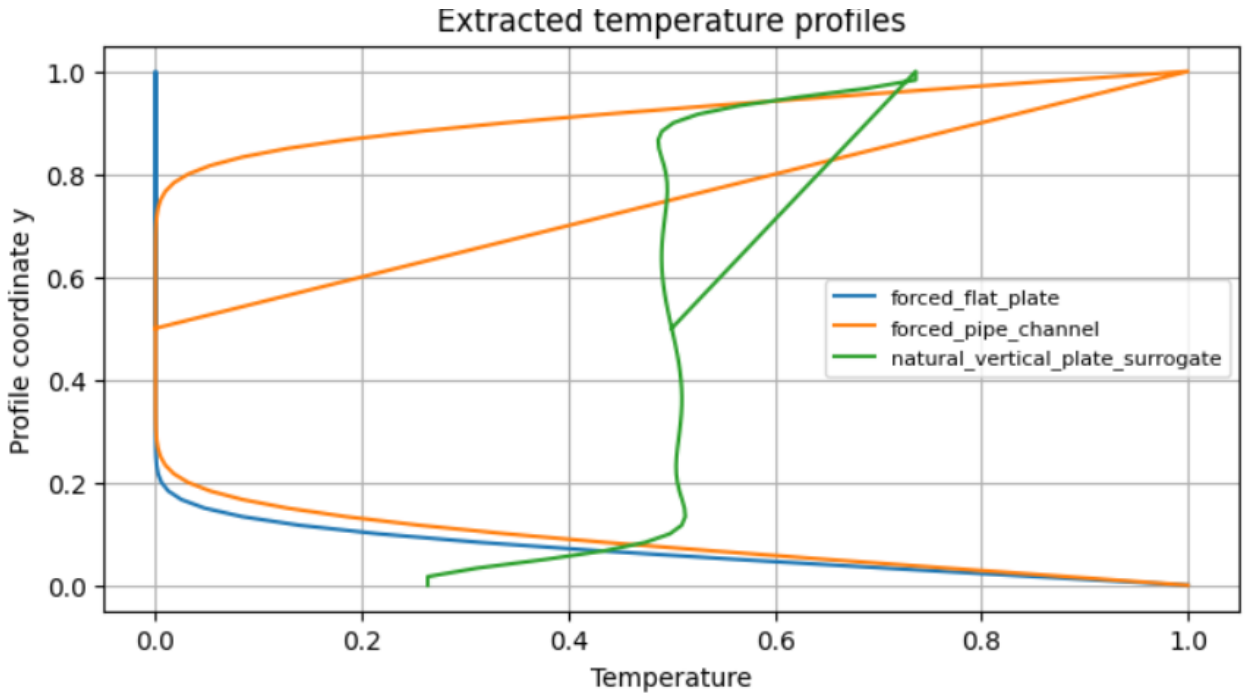
For Peclet number 200:

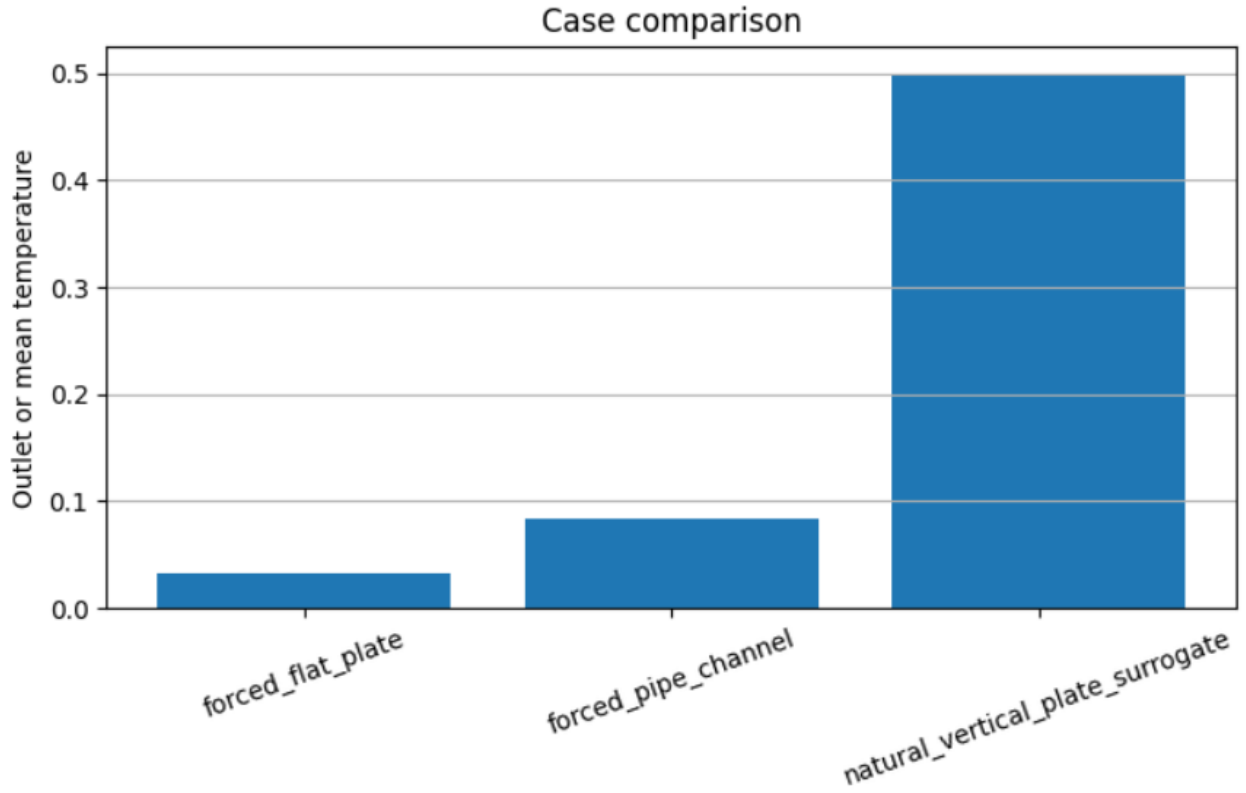


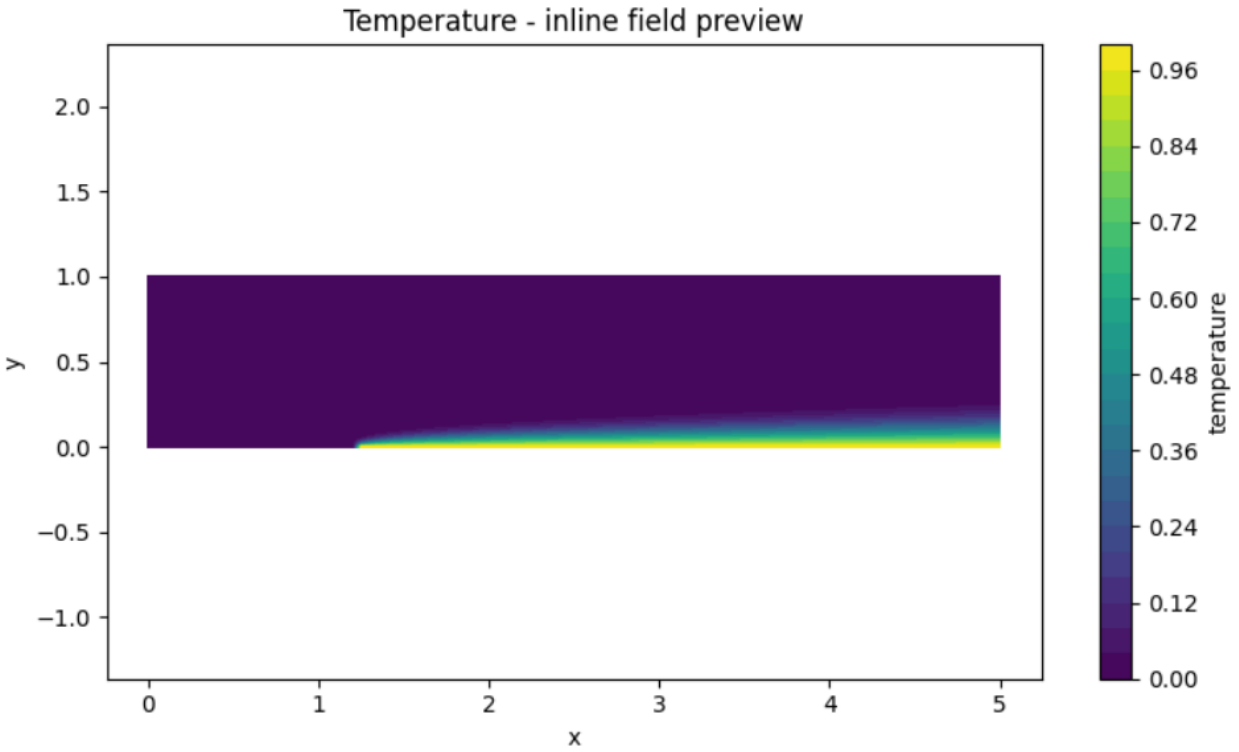

Case comparison
Outlet or mean temperature
0.0
0.1
0.2
0.3
0.4
0.5
forced_flat_plate
forced_pipe_channel
natural_vertical_plate_surrogate


Temperature - inline field preview
y
x
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
1
2
3
4
5
temperature
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00

For Peclet number 1000:







Results:

Parameter	**Pe = 50**	**Pe = 200**	**Pe = 1000**
Outlet / mean temperature: forced flat plate	≈ 0.24	≈ 0.097	≈ 0.032
Outlet / mean temperature: forced channel	≈ 0.565	≈ 0.237	≈ 0.083
Outlet / mean temperature: natural-convection surrogate	≈ 0.50	≈ 0.50	≈ 0.50
Scalar boundary-layer thickness at x = 5 (flat plate)	≈ 0.8–1.0	≈ 0.45	≈ 0.2
Channel centreline temperature (profile minimum)	≈ 0.18	≈ 0	≈ 0

Parameter	Pe = 50	Pe = 200	Pe = 1000
Field shape, forced cases	Thick, diffusive layer	Thinner layer	Thin layer hugging wall

Conclusion:

Scalar transport by upwind convection and central diffusion was compared for three prescribed velocity fields: a forced flat plate, a forced channel and a natural-convection surrogate with hot and cold vertical walls. The cases were run at Peclet numbers of 50, 200 and 1000 on a 65 × 65 grid. In every case the temperature field is bounded between the wall values, and the forced cases show the expected thermal boundary layer growing downstream of the start of the heated wall.

Effect of Peclet number on field shape and boundary-layer thickness: For the forced flat plate, the heated layer starts at about x = 1.25 and grows downstream. At Pe = 50 it is thick and diffuse, reaching about y = 0.8–1.0 at x = 5. At Pe = 200 it is about 0.45 thick, and at Pe = 1000 it is a thin layer of about 0.2 attached to the wall. The thickness falls by roughly a factor of 2 for each four- to five-fold increase in Pe, which is consistent with the usual scaling $\delta \propto Pe^{(-1/2)}$. As convection dominates over diffusion, heat is carried downstream before it can spread across the stream, so the temperature gradient at the wall becomes steeper.

Outlet or mean temperature: The outlet or mean temperature of the forced cases decreases strongly with Pe. For the flat plate it falls from about 0.24 to 0.097 to 0.032, and for the channel from about 0.565 to 0.237 to 0.083, at Pe = 50, 200 and 1000. The channel is warmer than the flat plate by a factor of about 2.4–2.6 at every Pe, because its two heated walls are close together and confine the heated fluid, while the flat plate is open on one side. The natural-convection surrogate stays at about 0.50 for all three Pe, since its hot and cold walls are symmetric and the recirculating flow redistributes heat between them, so its mean temperature does not depend on Pe.

Temperature profiles: In the channel, the centreline temperature falls from about 0.18 at Pe = 50 to nearly zero at Pe = 200 and 1000, and the two wall layers separate, leaving a cold core between them. At Pe = 50 the layers from the two walls are still merging. The natural-convection profile is different from the forced ones. It has a nearly uniform core near 0.5, with thin layers near the walls at Pe = 200 and 1000, and a more strongly curved profile at Pe = 50. This is the behaviour expected when a recirculating flow mixes the core and leaves the gradient in the wall layers.

Limitations of the figures: The wall flux, the final residual, the convergence history and the local Peclet number were requested in the observations but are not shown in the figures, so wall-flux and convergence comparisons cannot be made from the results here. The boundary-layer thicknesses and the mean temperatures above are read from the field plots and bar charts, so they are approximate. The extracted-profile plots also contain straight connecting segments (for example near y = 0.5), which look like joins between separately extracted profiles and not physical temperature distributions, so they were not used for quantitative values. In addition, the parameter

table lists Reynolds number, lid velocity, CFL, SOR and a tolerance of 1e-0.8, which appear to belong to a different (lid-driven) case and not to this scalar-transport study.

Overall, the three velocity fields give clearly different scalar transport. In the forced cases the thermal layer becomes thinner and the outlet or mean temperature falls as Pe increases, with the channel retaining more heat than the open flat plate. The natural-convection surrogate gives a mean temperature of about 0.5 at all Pe and has a mixed core, so it is controlled by the recirculation and not by the Peclet number. Reporting the wall flux, residual history and local Peclet number would complete the comparison of heat-transfer rate and convergence.